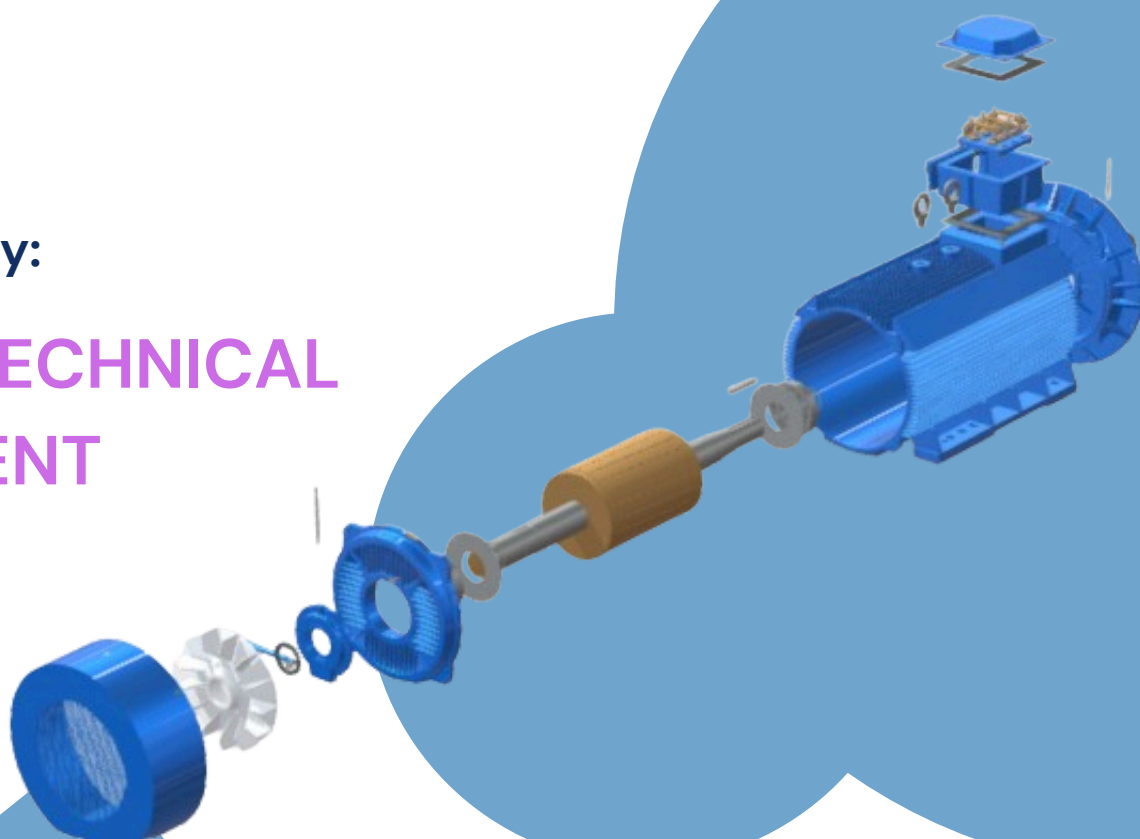




COMPENDIUM OF INDIAN STANDARDS ON ELECTRIC MOTORS

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INTRODUCTION

Electric motors are the workhorses of modern industry, powering a vast array of machines and processes. From the whirring fans in our homes to the colossal pumps in water treatment plants, motors play a vital role in our daily lives. As the demand for energy efficiency and environmental sustainability increases, optimizing motor performance and selecting the right motor for the job becomes crucial.

This compendium presents an organized overview of Indian Standards applicable to rotating electrical machines, developed by the Bureau of Indian Standards (BIS) and aligned with corresponding IEC standards to ensure international conformity. These standards encompass a wide spectrum—from fundamental performance and construction parameters to safety, testing, environmental conditions, and special applications.

Understanding these standards empowers engineers, procurement specialists, and facility managers to make informed decisions when selecting and using motors. By adhering to these standards, stakeholders can ensure the efficient operation of motors, leading to significant energy savings, reduced carbon footprint, and overall cost optimization.

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1 PRODUCT STANDARDS OF ELECTRIC MOTORS

1.1 CORE MOTORS PRODUCT STANDARDS

1. IS/IEC 60034-1 : 2022/ IEC 60034-1:2022 (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 1: Rating and Performance (Excluding Machines for Traction Vehicles)*

This is the core standard that defines the essential performance parameters and rating methods for rotating electrical machines. It covers general-purpose AC and DC machines and prescribes limits on temperature rise, output ratings, duty types, efficiency, and other performance metrics. It serves as the foundation for evaluating the design and operational compliance of motors and generators under standardized conditions.

2. IS 12615 : 2018/ IEC 60034-30-1 : 2014 (modified) (under revision)

Title: *Line-Operated Three-Phase a.c. Motors – Efficiency Classes and Performance Specification (IE Code)*

This Indian adaptation introduces **energy efficiency classifications (IE2 to IE4)** for three-phase induction motors, aligning with global practices. It specifies testing methods, losses, and performance benchmarks. This standard is key for implementing energy-efficient motor programs under Indian energy conservation initiatives.

3. IS 996:2009 (Under Revision)

Title: *Single Phase a.c. Induction Motors for General Purpose – Specification*

This standard covers the constructional, performance, and test requirements for **single-phase induction motors**, commonly used in domestic and light industrial applications. The standard includes duty types, ratings, starting methods, and mechanical design details. It is currently under revision to align with modern usage and energy efficiency norms.

1.2 HAND-HELD MOTOR-OPERATED ELECTRIC TOOLS STANDARDS

This section addresses **safety requirements for hand-held electric tools** that incorporate rotating electrical machines (motors). These standards are essential to protect users from hazards such as electric shock, mechanical injury, thermal effects, and exposure to vibration and noise. The standards follow a modular structure: **Part 1 provides general safety requirements**, while **Part 2 consists of tool-specific requirements** for drills, saws, grinders, hammers, and spray guns.

1. IS 4665 (Part 1):2023 / IEC 60745-1:2006 (Identical)

Title: *Hand-Held Motor-Operated Electric Tools – Safety – Part 1: General Requirements (Second Revision)*

This standard establishes **fundamental safety principles** applicable to all types of hand-held motor-operated electric tools, including insulation, grounding, protection against access to live parts, temperature rise limits, and mechanical robustness. It also specifies general test procedures for electrical, thermal, and mechanical safety, making it the base document for all tool-specific standards in the series.

2. IS 4665 (Part 2/Sec 1):2023 / IEC 60745-2-1 (Identical)

Title: *Part 2/Sec 1 – Particular Requirements for Drills and Impact Drills*

This section sets out **additional requirements and test methods** specifically for electric drills and impact drills. It addresses hazards due to chuck loosening, rotational forces (kickback), and drill bit ejection. Enhanced testing protocols ensure user safety during high-torque operation and under locked-rotor conditions.

3. IS 4665 (Part 2/Sec 2):2023 / IEC 60745-2-2 (Identical)

Title: *Part 2/Sec 2 – Particular Requirements for Screwdrivers and Impact Wrenches*

This standard provides **safety provisions for tools designed to apply torque**, such as electric screwdrivers and impact wrenches. It focuses on torque control mechanisms, durability of moving parts, and user safety under repetitive impact conditions.

4. IS 4665 (Part 2/Sec 3):2023 / IEC 60745-2-3 (Identical)

Title: *Part 2/Sec 3 – Particular Requirements for Grinders, Polishers, and Disk-Type Sanders*

It specifies **additional mechanical safety and guarding requirements** for abrasive tools with high-speed rotating parts. Topics covered include wheel burst protection, spindle locking, thermal protection, and balance testing for rotating disks.

5. IS 4665 (Part 2/Sec 5):2023 / IEC 60745-2-5 (Identical)

Title: *Part 2/Sec 5 – Particular Requirements for Circular Saws*

This section includes **special safety provisions for hand-held circular saws**, focusing on blade guarding, anti-kickback design, and stopping time after switch-off. It also includes stability and insulation checks for saws operating under load.

6. IS 4665 (Part 2/Sec 6):2023 / IEC 60745-2-6 (Identical)

Title: *Part 2/Sec 6 – Particular Requirements for Hammers*

This part applies to rotary and demolition hammers. It emphasizes **vibration safety, bit retention, and structural integrity** during percussive operation. The standard ensures the tools are capable of withstanding repeated impact without user injury.

7. IS 4665 (Part 2/Sec 7):2023 / IEC 60745-2-7 (Identical)

Title: *Part 2/Sec 7 – Particular Requirements for Spray Guns for Non-Flammable Liquids*

This section covers spray guns used in painting or coating applications. It defines **requirements for liquid sealing, nozzle protection, and electrical isolation**, ensuring safe usage especially in industrial settings.

1.3 AGRICULTURAL MOTOR STANDARDS

This section includes Indian Standards developed for **rotating electrical machines used in agricultural applications**, primarily focusing on motors that drive pumps for irrigation and water lifting. These motors are specially designed for **reliable operation in rural environments**, often under fluctuating voltage, dusty conditions, and limited maintenance. The standards ensure energy-efficient, durable, and field-suitable motors for centrifugal and submersible pump sets.

1. IS 9283:2024

Title: *Line Operated A.C. Motors for Submersible Pump Sets – Specification (Third Revision)*

This standard specifies the design, construction, and performance of **AC motors used for submersible pump sets**, which are deployed in deep bore wells for irrigation. It covers both single-phase and three-phase motors, including oil-filled and water-filled types. Key features include **slim body construction, moisture sealing, and higher thrust bearing capacities** to handle vertical mounting and hydrostatic pressure. The standard is aligned with modern energy-efficiency norms and ensures operational safety under submerged conditions.

2. IS 7538:1996

Title: *Three-Phase Squirrel Cage Induction Motors for Centrifugal Pumps for Agricultural Application – Specification (First Revision)*

This standard lays down the requirements for **three-phase squirrel cage induction motors** used in driving centrifugal water pumps in agriculture. It includes specifications for motor rating, voltage range (typically for rural feeder conditions), design for high ambient temperatures, and overload capacities. Mechanical features such as foot mounting, terminal box orientation, and shaft dimensions are also covered. The focus is on **ruggedness, reliability, and simplicity** of design.

3. IS 14582:2021

Title: *Single-Phase Small AC Electric Motors for Centrifugal Pumps for Agricultural Applications*

This standard focuses on **small-capacity, single-phase induction motors** typically used with shallow well or surface-mounted centrifugal pumps. It includes performance requirements such as starting torque, voltage tolerance, capacitor specifications, and thermal protection. These motors are crucial in low-load, small-farm setups, and are designed for **economical operation and easy field serviceability**.

1.4 SPECIFIC APPLICATION MOTORS

1. IS 14578:1999 (Under Revision)

Title: *Three-Phase Induction Motors for Use in Nuclear Power Plants – Specification*

This is a specialized standard specifying design, construction, and testing requirements for induction motors used in nuclear power installations. These motors must meet enhanced safety, reliability, and environmental qualification standards. The document ensures suitability under seismic, thermal, and radiation stress conditions.

2. IS 8151:2024

Title: *Single-Speed Three-Phase Induction Motors for Driving Lifts – Specification (First Revision)*

IS 8151:2024 specifies the performance and constructional requirements for **single-speed three-phase induction motors used in lift (elevator) applications**. These motors are specially designed to deliver **high starting torque, smooth acceleration, and consistent performance** under frequent start-stop duty cycles common in vertical transport systems. The standard outlines parameters for **duty type, thermal classification, starting current, torque characteristics**, and mechanical construction to ensure safety, reliability, and efficient operation. It also addresses vibration limits, insulation, and protection classes, ensuring compatibility with elevator gear mechanisms and compliance with safety codes.

3. IS 18073:2023

Title: *Electric Traction Motor – Performance and Functional Requirements*

IS 18073:2023 (published under TED 27 committee) establishes comprehensive performance and functional requirements for electric traction motors. The standard encompasses critical aspects such as safety, environmental suitability, electromagnetic compatibility (EMC), and durability. By aligning with international best practices, IS 18073:2023 aims to ensure the reliability, safety, and efficiency of electric traction systems, thereby supporting the advancement of electric mobility in India.

1.5 BRUSH STANDARDS

This section focuses on standards related to **brushes and brush-holders** used in rotating electrical machines—mainly in DC motors, slip ring induction motors, and some synchronous machines. These components are critical for reliable current transfer between stationary and rotating parts. The standards ensure consistency in **design, performance, contact pressure, material compatibility**, and safe operation of brushes under different loading conditions.

1. IS 12642:1989 / IEC 60778 (Identical)

Title: *Brush-Holders for Slip Rings Group R, Type RA – Specification*

This standard specifies the **dimensional and constructional requirements** for brush-holders of type RA used with slip rings in rotating machines. It ensures compatibility between brush-holders and brushes, and defines mounting arrangements, clearances, and material guidelines. This standard is especially relevant for machines where slip ring current transmission is required—such as wound rotor induction motors and synchronous generators.

2. IS 13466:1992 (Under Revision)

Title: *Brushes for Electrical Machines – Specification*

IS 13466 provides a **comprehensive specification for carbon brushes** used in rotating machines. It includes classifications based on brush material (e.g., electrographitic, resin-bonded), physical dimensions, electrical characteristics, and performance under mechanical and thermal stress. The standard also defines testing methods for brush wear, temperature rise, and contact drop. It is currently under revision to incorporate new materials and usage patterns.

3. IS 13525:1992 (Under Revision)

Title: *Flexible Conductors for Carbon Brushes – Specification*

This standard specifies requirements for **flexible braided conductors** (pigtailed) used to connect carbon brushes to current-carrying terminals. It covers aspects such as conductor material, strand configuration, current capacity, and resistance to mechanical fatigue. Proper design ensures uninterrupted power transfer and minimizes vibration-induced failures. This standard supports high-reliability motor applications.

2 TEST METHOD STANDARDS

2.1 BASIC TEST METHOD STANDARDS

1. IS/IEC 60034-2-1 : 2024/IEC 60034-2-1:2024 (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 2: Determining Losses and Efficiency from Tests – Section 1: Standard Methods (Excluding Machines for Traction Vehicles)*

IS/IEC 60034-2-1:2024 specifies standardized methods for determining the total losses and efficiency of rotating electrical machines (excluding traction motors) through testing. It defines direct and indirect test methods suitable for a wide range of machine types and sizes. The standard includes:

- Input-output method (direct efficiency measurement)
- Summation of losses with load test using segregated losses and stray load loss from residual loss at rated load
- Summation of losses with load test using segregated losses and calculated stray load losses
- Summation of losses with load test using segregated losses and fixed stray load loss (0.5% of input)
- Calorimetric method (for machines with significant cooling losses)

These methods are intended for accuracy classes 0.1 to 1.0, and the standard includes detailed procedures for loss separation (e.g., stator copper, rotor copper, iron, mechanical, and stray load losses), corrections for temperature and frequency, and harmonized conditions for comparison of efficiencies. The guidance ensures reliable, reproducible, and comparable efficiency data for product evaluation, compliance, and labeling.

2. IS/IEC 60034-18-42) : 2017/ IEC 60034-18-42:2017+AMD1:2020 CSV (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 18: Partial Discharge Resistant Electrical Insulation Systems (Type II) Fed from Voltage Converters – Section 42: Qualification Tests*

This standard defines qualification tests for **Type II insulation systems**, which are capable of withstanding partial discharges under repetitive voltage stress typical of PWM converters. These systems are designed for improved lifetime performance in demanding converter-fed machine applications. The tests focus on thermal endurance, electrical stress resilience, and long-term partial discharge behavior.

3. IS/IEC 60034-18-41 : 2014/ (IEC 60034-18-41:2014+AMD1:2019 CSV) (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 18: Partial Discharge Free Electrical Insulation Systems – Section 41: Type I Systems Fed from Voltage Converters – Qualification and Quality Control Tests*

This specialized standard is concerned with the insulation systems used in rotating machines supplied by modern voltage converters (like variable frequency drives). It lays out tests to qualify

insulation systems that are resistant to partial discharges induced by fast switching waveforms. The focus is on Type I insulation systems and includes both qualification and ongoing quality control measures to ensure long-term reliability.

4 IS/IEC 60034-27-4 : 2018/IEC 60034-27-4 : 2018 (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 27: Winding Insulation – Section 4: Measurement of Insulation Resistance and Polarization Index*

This standard lays out procedures for measuring the **insulation resistance (IR)** and **polarization index (PI)** of machine windings, key indicators of insulation health. These tests are simple but powerful diagnostic tools, often used in preventive maintenance. They help assess insulation degradation due to moisture, contamination, or thermal aging.

5. IS 4029:2010 (Under Revision)

Title: *Guide for Testing Three-Phase Induction Motors (First Revision)*

This standard provides detailed guidance on the **type, routine, and special tests** for three-phase induction motors. It includes insulation resistance, winding resistance, no-load and full-load tests, vibration, noise, and heating tests. The objective is to verify compliance with specifications and operational safety before dispatch or after repair. It is under revision to reflect modern motor designs and testing equipment capabilities.

6. IS 9320:2025

Title: *Guide for Testing Direct Current (DC) Machines (First Revision)*

This updated guide outlines test procedures for **DC motors and generators**, including commutation, insulation, armature reaction, temperature rise, and efficiency tests. It ensures DC machines meet performance and safety requirements as per application demands in traction, industrial drives, or process automation.

7. IS 7572:1974 (Under Revision)

Title: *Guide for Testing Single-Phase AC and Universal Motors*

This standard specifies procedures for testing small single-phase AC motors and **universal motors** used in household appliances and tools. The guide includes performance checks, dielectric strength, efficiency, and thermal limits. It is under revision to reflect modern motor designs and testing equipment capabilities.

8. IS 12075:2024 / IEC 60034-14:2018 (Modified)

Title: *Mechanical Vibration of Rotating Electrical Machines with Shaft Heights 56 mm and Higher – Measurement, Evaluation and Limits of Vibration Severity*

This standard details how to measure and assess **vibration severity** in machines with medium

and large shaft heights. Acceptable limits are prescribed to prevent mechanical damage, premature bearing failure, and structural resonance. It's widely used during factory acceptance testing and commissioning.

2.2 NOISE MEASUREMENT STANDARDS

This category covers standards for the **measurement, declaration, and verification of noise emissions** from rotating electrical machines and industrial equipment. Accurate noise data is crucial for ensuring compliance with environmental norms, workplace safety, and acoustic design criteria. These standards use statistical and direct measurement methods to determine noise levels for individual machines or batches.

1. IS 13937 (Part 1):1994 / ISO 7574-1 (Identical)

Title: *Statistical Methods for Determining and Verifying Stated Noise Emission Values of Machinery and Equipment – Part 1: General Considerations and Definitions*

This standard lays the foundation for the entire series by defining the **terminology, statistical approaches, and basic principles** involved in declaring and verifying noise emission values. It introduces key concepts like statistical uncertainty, measurement repeatability, and types of declared values, establishing a consistent framework for all subsequent parts.

2. IS 13937 (Part 2):1994 / ISO 7574-2 (Identical)

Title: *Statistical Methods – Part 2: Methods for Stated Values for Individual Machines*

This part focuses on the **noise measurement of individual machines**, providing procedures to determine and verify the sound power level emitted. It includes confidence intervals and verification criteria, especially for machines produced in small numbers or custom applications.

3. IS 13937 (Part 3):1994 / ISO 7574-3 (Identical)

Title: *Statistical Methods – Part 3: Simple (Transition) Method for Stated Values for Batches of Machines*

Designed for **batch-produced machines**, this method provides a simplified statistical approach for manufacturers to determine noise emission values. It acts as a transition method between basic and rigorous statistical techniques, suitable for medium production volumes.

4. IS 13937 (Part 4):1994 / ISO 7574-4 (Identical)

Title: *Statistical Methods – Part 4: Methods for Stated Values for Batches of Machines*

This part provides **detailed statistical procedures** to determine and verify stated noise values for **large batches** of identical machines. It is especially useful in factory acceptance testing, product certification, and conformity assessments, ensuring reproducibility and traceability of acoustic data.

5. IS 12998:2024 / ISO 1680-1:2013 (Identical)

Title: *Acoustics – Test Code for the Measurement of Airborne Noise Emitted by Rotating Electrical Machines*

This standard defines **practical methods for measuring airborne sound** power levels emitted by rotating electrical machines under standardized operating conditions. It includes specifications for test setups, environmental corrections, background noise adjustment, and instrumentation calibration. It ensures uniformity in reporting and facilitates comparison across manufacturers and models.

11. IS 12065 : 2025/ IEC 60034-9:2021 (modified)

Title: *Permissible Limits of Noise Levels for Rotating Electrical Machines*

This standard defines measurement procedures and permissible noise limits for electrical machines under standard operating conditions. It accounts for sound pressure levels emitted by machines and supports quieter machine design, workplace safety, and noise compliance in sensitive applications.

3 DIMENSIONS AND SYMBOLS STANDARDS

Dimensional standardization ensures **interchangeability, uniformity in design, and compatibility** of motors with mechanical equipment like pumps, compressors, and gearboxes. The following Indian standards—many aligned with international norms—cover dimensional aspects such as frame sizes, mounting types, shaft extensions, and flange details.

1. IS 1231:2019

Title: *Dimensions and Output Series of Foot-Mounted Induction Motors — Frame Numbers 56 to 315 L (Fourth Revision)*

This standard specifies **frame sizes, mounting dimensions, and shaft details** for foot-mounted squirrel cage induction motors. It covers frame sizes from 56 to 315L, which are commonly used in a wide variety of industrial applications. This helps ensure dimensional compatibility across manufacturers and facilitates equipment replacement and maintenance.

2. IS 2223:1983 (Under Revision)

Title: *Dimensions of Flange-Mounted AC Induction Motors*

IS 2223 covers the **flange mounting dimensions (Type B5 and B14)** of AC induction motors. These flange-mounted motors are often used in compact applications like gearboxes or pumps where alignment and space are critical. Although dated, it has remained an important reference until its upcoming revision to align with IEC 60072.

3. IS 2254:1985 (Under Revision)

Title: *Dimensions of Vertical Shaft Motors for Pumps (Second Revision)*

This standard defines the **dimensional requirements for vertically mounted motors**, particularly those used in pump applications. It includes specifications for shaft dimensions, bearing arrangements, and flange interfaces to ensure seamless integration with pump units. Revision efforts aim to align it more closely with contemporary mechanical and hydraulic pump designs.

4. IS 8223:1999 / IEC 60072-2 (Identical)

Title: *Dimensions and Output Series for Rotating Electrical Machines (First Revision)*

This standard provides a generalized **output and frame size series** for rotating machines, regardless of specific type. It is harmonized with IEC 60072-2 and ensures standardized motor sizes across different machine types (AC/DC, foot/flange mounted). It aids in rational design practices and cross-brand compatibility in industrial setups.

5. IS 15999 (Part 7) : 2024/ IEC 60034-7:2020 (identical)

Title: *Rotating Electrical Machines – Part 7: Classification of Types of Construction, Mounting Arrangements, and Terminal Box Position (IM Code)*

This standard provides a systematic coding system (IM Code) for identifying the mounting arrangements, structural designs, and terminal box orientations of rotating electrical machines. It enables standardized identification across manufacturers and users, simplifying installation and interchangeability across applications. It's essential for mechanical integration in plant design.

6. IS/IEC 60034-8 : 2014/IEC 60034-8 : 2014 (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 8: Terminal Markings and Direction of Rotation*

This document standardizes the marking of terminals and ensures clarity on the direction of rotation for three-phase and single-phase machines. Correct terminal marking is crucial for safe installation, phase sequencing, and ensuring desired rotational direction. This standard promotes consistency across equipment and minimizes wiring errors.

7. IS 6362:1995 / IEC 60034-6 (Identical)

Title: *Designation of Methods of Cooling for Rotating Electrical Machines*

This standard establishes a coded system for identifying **cooling methods** used in electrical machines, such as IC01 (open air), IC411 (self-ventilated with frame-mounted cooler), etc. Proper cooling classification ensures safe thermal performance and helps in the selection of machines for different environmental and duty conditions.

4 GUIDES AND CODE OF PRACTICE STANDARDS

This category includes **testing guides and operational best practices** for various types of rotating machines. These standards are essential for ensuring **correct testing, safe operation, reliability, and long service life** of electrical motors and generators. They are widely used by testing labs, quality assurance engineers, maintenance personnel, and commissioning teams.

1. IS/IEC 60034-5 : 2020 / IEC 60034-5:2020 (Identical under single numbering)

Title: *Rotating Electrical Machines – Part 5: Degrees of Protection Provided by the Integral Design (IP Code) – Classification*

This standard specifies the enclosure protection levels against the ingress of solid objects and water (IP ratings) as designed into the rotating machine itself. It helps in categorizing machines based on their environmental protection capability, which is critical for specifying machines for industrial, outdoor, or hazardous environments. The IP classification aligns with broader IEC 60529 coding.

2. IS 900:2019

Title: *Code of Practice for Storage, Installation, and Maintenance of Induction Motors (Third Revision)*

IS 900 provides structured procedures for the **handling, storage, installation, alignment, and preventive maintenance** of induction motors. It helps users ensure motors retain their performance over time, particularly in industrial environments where factors like humidity, vibration, and dust can degrade equipment prematurely.

3. IS 15429:2004

Title: *Storage, Installation, and Maintenance of DC Motors – Code of Practice*

Similar to IS 900 but focused on **DC motors**, this standard provides guidelines for long-term motor preservation, shaft locking, bearing lubrication, and installation practices specific to DC machinery. It's particularly relevant in heavy-duty or mission-critical systems like rolling mills or defense systems where downtime is unacceptable.

4. IS/IEC 60034-23:2019 (under print)

Title: *Rotating Electrical Machines – Part 23: Repair, Overhaul, and Reclamation*

This standard provides **comprehensive guidelines for the repair, refurbishment, and overhaul** of rotating electrical machines, including AC and DC types. It emphasizes maintaining or restoring original design performance, safety, and reliability during repair. The standard covers aspects such as disassembly, winding replacement, insulation checks, mechanical tolerances, and testing post-repair. It also promotes traceability and quality control during reclamation, helping reduce life-cycle costs while maintaining machine integrity.

5. IS/IEC TS 60034-25:2022 (under print)

Title: *Rotating Electrical Machines – Part 25: AC Electrical Machines Used in Power Drive Systems – Application Guide*

This technical specification serves as an **application guide for AC machines used with power electronic converters** in Power Drive Systems (PDS). It addresses machine behavior under voltage modulation, insulation stress, harmonic heating, and bearing currents resulting from converter-fed operation. It also includes recommendations for selecting compatible motor-converter combinations, cooling needs, and derating factors. This guide is essential for ensuring reliable operation in industrial automation, robotics, and variable-speed drive applications.

6. IS 14195:1994 / IEC 61015 (Identical)

Title: *Brush-Holders for Electrical Machines – Guide to the Measurement of the Static Thrust Applied to Brushes*

This guide describes **methods for measuring the static pressure** (or thrust) exerted by brush-holders on brushes. Correct contact pressure is essential for efficient current transfer, reduced wear, and preventing arcing or overheating. The standard helps maintain optimum brush life and motor performance by specifying mechanical and measurement techniques to verify thrust settings.

7. IS 13529:2021 / IEC 60034-26:2006 (Modified)

Title: *Guide on Effects of Unbalanced Voltages on the Performance of Three-Phase Cage Induction Motors (First Revision)*

IS 13529:2021 provides a comprehensive guide on the **impact of voltage unbalance** on the performance, efficiency, and thermal behavior of **three-phase squirrel cage induction motors**. Voltage unbalance, even when minor, can lead to **increased losses, reduced torque output, overheating**, and premature insulation failure. This standard defines acceptable limits of unbalance, explains methods to quantify it (e.g., percentage voltage unbalance), and provides **derating factors** to adjust motor loading under such conditions. It serves as a valuable reference for motor users and designers to **mitigate risks**, ensure motor longevity, and maintain energy-efficient operation in real-world, often imperfect, supply conditions.